

Midterm 1 – Study Notes

(Weeks 1, 2 & 3)

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Week 1 – Propositional Logic & Reasoning

Reasoning Types

Inductive Reasoning

Drawing general conclusions from a limited set of observations.

Cannot guarantee correctness.

Deductive Reasoning

Using logic to draw conclusions from statements we accept as true.

Conclusions are certain if premises are true.

Conditional Statements

Given a conditional $p \rightarrow q$ (“If p , then q ”):

Name	Form	Example
Original	$p \rightarrow q$	If it rains, the grass is wet.
Converse	$q \rightarrow p$	If the grass is wet, it rained.
Contrapositive	$\neg q \rightarrow \neg p$	If the grass is not wet, it did not rain.
Inverse	$\neg p \rightarrow \neg q$	If it does not rain, the grass is not wet.

Key fact: A conditional and its contrapositive are logically equivalent. The converse and inverse are logically equivalent to each other, but **not** to the original.

Logical Connectives

Negation	$\neg p$	NOT p
Conjunction	$p \wedge q$	p AND q
Disjunction	$p \vee q$	p OR q
Implication	$p \rightarrow q$	If p then q
Biconditional	$p \leftrightarrow q$	p if and only if q

Truth Table Reference

p	q	$\neg p$	$p \wedge q$	$p \vee q$	$p \rightarrow q$	$p \leftrightarrow q$
T	T	F	T	T	T	T
T	F	F	F	T	F	F
F	T	T	F	T	T	F
F	F	T	F	F	T	T

Important: $p \rightarrow q$ is only false when p is true and q is false.

Algebraic Proof: $(-a)(-b) = ab$

$$0 = 0 \cdot 0$$

$$= (a - a)(b - b)$$

$$= ab - ab - ab + (-a)(-b)$$

$$ab = (-a)(-b)$$

Property of zeros

Definition of inverses

FOIL

Add ab to both sides, simplify

Week 2 – Logical Equivalences, Quantifiers & Bitwise Ops

Key Logical Equivalences (Chart 7, Rosen p.28)

1. **RBI (Really Big Idea):** $p \rightarrow q \equiv \neg p \vee q$
2. **Contrapositive:** $p \rightarrow q \equiv \neg q \rightarrow \neg p$
3. $p \vee q \equiv \neg p \rightarrow q$
4. $p \wedge q \equiv \neg(p \rightarrow \neg q)$
5. $\neg(p \rightarrow q) \equiv p \wedge \neg q$

Proof strategy: Most of 2–5 follow from applying RBI and substitution. RBI converts conditionals into disjunctions (and vice versa), which is the foundation for the rest.

Negation of Conditionals

From equivalence #5 above:

$$\neg(p \rightarrow q) \equiv p \wedge \neg q$$

In words: the negation of “if p then q ” is “ p is true **and** q is false.”

Why: Apply RBI then De Morgan’s:

$$\begin{aligned}\neg(p \rightarrow q) &\equiv \neg(\neg p \vee q) && \text{RBI} \\ &\equiv p \wedge \neg q && \text{De Morgan's}\end{aligned}$$

This is useful for **proof by contradiction:** to assume a conditional is false, assume the hypothesis is true and the conclusion is false.

Important Laws (Table 6, Rosen p.27)

De Morgan’s Laws:

$$\begin{aligned}\neg(p \wedge q) &\equiv \neg p \vee \neg q \\ \neg(p \vee q) &\equiv \neg p \wedge \neg q\end{aligned}$$

Distributive Laws:

$$\begin{aligned}p \vee (q \wedge r) &\equiv (p \vee q) \wedge (p \vee r) \\ p \wedge (q \vee r) &\equiv (p \wedge q) \vee (p \wedge r)\end{aligned}$$

Other Essential Laws:

$$\begin{aligned}\text{Double Negation:} & \quad \neg(\neg p) \equiv p \\ \text{Commutative:} & \quad p \wedge q \equiv q \wedge p, \quad p \vee q \equiv q \vee p \\ \text{Associative:} & \quad (p \wedge q) \wedge r \equiv p \wedge (q \wedge r), \quad \text{same for } \vee \\ \text{Identity:} & \quad p \wedge T \equiv p, \quad p \vee F \equiv p \\ \text{Domination:} & \quad p \vee T \equiv T, \quad p \wedge F \equiv F \\ \text{Idempotent:} & \quad p \wedge p \equiv p, \quad p \vee p \equiv p \\ \text{Absorption:} & \quad p \vee (p \wedge q) \equiv p, \quad p \wedge (p \vee q) \equiv p\end{aligned}$$

Logical FOIL

$$(a \wedge b) \vee (c \wedge d) \equiv (a \vee c) \wedge (b \vee c) \wedge (a \vee d) \wedge (b \vee d)$$

Proved by applying distributive law twice (factor using substitution $S = a \wedge b$).

Quantifiers & Predicates

Universal $\forall x P(x)$ “ $P(x)$ is true for **all** x ”

Existential $\exists x P(x)$ “There exists an x such that $P(x)$ is true”

Negation of Quantifiers:

$$\neg \forall x P(x) \equiv \exists x \neg P(x)$$

$$\neg \exists x P(x) \equiv \forall x \neg P(x)$$

Nested quantifiers: Negate from left to right, flipping each quantifier:

$$\neg \forall x \forall y P(x, y) \equiv \exists x \exists y \neg P(x, y)$$

$$\neg \forall y \exists x P(x, y) \equiv \exists y \forall x \neg P(x, y)$$

When a connective appears inside, apply De Morgan’s after flipping quantifiers:

$$\neg \forall y \forall x (P(x, y) \vee Q(x, y)) \equiv \exists y \exists x (\neg P(x, y) \wedge \neg Q(x, y))$$

Bitwise Operations

Given bit strings x and y , apply bit-by-bit:

x	y	AND ($x \wedge y$)	OR ($x \vee y$)	XOR ($x \oplus y$)
0	0	0	0	0
0	1	0	1	1
1	0	0	1	1
1	1	1	1	0

XOR note: Returns 1 when bits differ, 0 when they are the same.

Week 3 – Proof Techniques & Even/Odd Proofs

Definitions

Even integer: n is even if there exists an integer k such that $n = 2k$

Odd integer: n is odd if there exists an integer k such that $n = 2k + 1$

Proof Techniques

1. **Direct Proof** ($p \rightarrow q$): Assume p is true, show q follows.

2. **Proof by Contrapositive** ($\neg q \rightarrow \neg p$): Instead of proving $p \rightarrow q$, prove the logically equivalent $\neg q \rightarrow \neg p$. Useful when starting from p doesn't give a clean setup (e.g. dividing by 3 breaks the integer structure).

3. **Proof by Contradiction:** Assume $\neg(p \rightarrow q) \equiv p \wedge \neg q$, then derive a logical contradiction. Conclude the original statement must be true.

4. **Proof of Equivalence** (biconditional $p \leftrightarrow q$): Must prove **both** directions:

$$(p \leftrightarrow q) \equiv (p \rightarrow q) \wedge (q \rightarrow p)$$

Each direction can use any technique (direct, contrapositive, etc.).

Proof Toolkit (for Even/Odd Proofs)

1. **Substitution:** Replace a variable with its definition (e.g. $n = 2k + 1$).

2. **Closure of integers:** The sum, difference, or product of integers is an integer. This justifies renaming expressions like $d + h$ or $2jk + j + k$ as a single integer variable.

3. **Factoring:** Pull out a factor of 2 to match the even/odd definition.

4. **FOIL:** Expand products like $(2j + 1)(2k + 1) = 4jk + 2j + 2k + 1$.

5. **Use distinct variables:** When two values are both even (or both odd), use **different** variables for each (e.g. $A = 2d$, $B = 2h$, not both using d).

Warm-Up Theorem Results (from HW3)

THM	Statement	Result
1	even + even	even
2	even + odd	odd
3	odd + odd	even
4	even $\times$ even	even
5	even $\times$ odd	even
6	odd $\times$ odd	odd

Note: These results can be *used* in later proofs (e.g. “by Theorem 6, the product of two odd integers is odd”) vs. re-deriving using the *techniques*.

Example: Direct Proof (THM 6 – odd $\times$ odd is odd)

Let $A = 2j + 1$, where j is an integer def. of odd

Let $B = 2k + 1$, where k is an integer def. of odd

$C = A \cdot B$ premise

$= (2j + 1)(2k + 1)$ substitution

$= 4jk + 2j + 2k + 1$ FOIL

$= 2(2jk + j + k) + 1$ factoring

Let $h = 2jk + j + k$, then h is an integer closure of integers

$= 2h + 1$ substitution

- $C = 2h + 1$, so by definition C is odd. □

Example: Proof by Contrapositive

Prove: If $3n + 7$ is odd, then n is even.

Why not direct? Starting with $3n + 7 = 2k + 1$ gives $n = \frac{2k-6}{3}$, and dividing by 3 prevents cleanly factoring out a 2.

Contrapositive: If n is odd, then $3n + 7$ is even.

Let $n = 2a + 1$, where a is an integer def. of odd

$$3n + 7 = 3(2a + 1) + 7 \quad \text{substitution}$$

$$= 6a + 3 + 7 \quad \text{distribution}$$

$$= 6a + 10 \quad \text{addition}$$

$$= 2(3a + 5) \quad \text{factoring}$$

Let $b = 3a + 5$, then b is an integer closure of integers

$$= 2b \quad \text{substitution}$$

- $3n + 7 = 2b$, so it is even. By contrapositive, if $3n + 7$ is odd then n is even. □

Example: Proof by Contradiction

Prove: There is no largest integer.

Assume for contradiction that there exists an integer z such that $z > n$ for all integers n .

Let $n = z + 1$. Since z is an integer, $z + 1$ is also an integer (closure). But then $z > z + 1$, which contradicts the properties of integer addition. □

Logarithm Review (Jacobs, Ch. 4)

Definition

A **logarithm** is an exponent. If $y = b^x$, then $x = \log_b y$.

Binary logarithm (base 2): The exponent of 2 that produces the number.
Example: $\log_2 8 = 3$ because $8 = 2^3$.

Decimal logarithm (base 10): The exponent of 10 that produces the number.
Example: $\log_{10} 1,000,000 = 6$ because $1,000,000 = 10^6$.

Three Laws of Logarithms

These hold for **any** base.

Operation	Numbers	Logarithms
Multiplication	$a \times b$	$\log(a \times b) = \log a + \log b$
Division	$a \div b$	$\log(a \div b) = \log a - \log b$
Powers	a^n	$\log(a^n) = n \cdot \log a$

In words:

- **Multiplication** of numbers $\longleftrightarrow$ **Addition** of their logarithms
- **Division** of numbers $\longleftrightarrow$ **Subtraction** of their logarithms
- **Raising to a power** $\longleftrightarrow$ **Multiplying** the logarithm by the exponent

Binary Logarithm Table (Base 2)

Number	1	2	4	8	16	32	64	128	256	512	1024
$\log_2$	0	1	2	3	4	5	6	7	8	9	10

Examples using the table:

$$\begin{aligned} 8 \times 16 &= 128 & \log_2 8 + \log_2 16 &= 3 + 4 = 7 = \log_2 128 \\ 512 \div 64 &= 8 & \log_2 512 - \log_2 64 &= 9 - 6 = 3 = \log_2 8 \\ 8^4 &= 4096 & 4 \cdot \log_2 8 &= 4 \times 3 = 12 = \log_2 4096 \end{aligned}$$

Decimal Logarithm Table (Base 10)

Number	1	2	3	4	5	6	7	8	9	10
$\log_{10}$	0	0.30	0.48	0.60	0.70	0.78	0.85	0.90	0.95	1

Number	10^1	10^2	10^3	10^4	10^5	10^6	10^7	10^8	10^9
$\log_{10}$	1	2	3	4	5	6	7	8	9

Logarithms with Scientific Notation

To find the logarithm of a number in scientific notation, add the log of the coefficient to the log of the power of 10:

$$\log(a \times 10^n) = \log a + \log 10^n = \log a + n$$

Example: $\log(2 \times 10^5) = \log 2 + 5 = 0.30 + 5 = 5.30$

Works in reverse too — given a logarithm, split into integer + decimal parts:

Example: $\log = 7.6 \Rightarrow 0.60 + 7$. From the table, $0.60 = \log 4$ and $7 = \log 10^7$, so the number is 4×10^7 .

Approximating Large Powers

Use the power rule to estimate values without a calculator:

Example: Find 9^9 .

$$\log 9 = 0.954$$

$$\log(9^9) = 9 \times 0.954 = 8.586$$

$$8.586 = 0.586 + 8$$

So $9^9 \approx 3.85 \times 10^8 = 385,000,000$ (a 9-digit number).